

SURYAPRABHA PJ

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SUMMARY

Computer Science undergraduate specializing in Data Science, Machine Learning, and Computer Vision with hands-on experience building real-time AI systems and predictive models. Passionate about transforming raw data into actionable insights that enhance accessibility and decision-making.

TECHNICAL SKILLS

Programming Languages: Python, Java

Data Science & Machine Learning: Data Analysis, Data Preprocessing, Feature Engineering, Machine Learning Algorithms, Deep Learning, Model Evaluation

Libraries & Frameworks: Pandas, NumPy, Scikit-learn, TensorFlow, Matplotlib, Seaborn

Data Visualization & BI Tools: Power BI, Microsoft Excel, Matplotlib, Seaborn

Databases & Web: MySQL, MongoDB, HTML, CSS

EDUCATION

Ramco Institute of Technology, Rajapalayam

Sep 2023 – Jul 2027

B.E in Computer Science and Engineering – 8.01 CGPA

PROJECTS

Eye Gaze Controlled Assistive Technology: [GitHub](#)

Mar 2025 - Present

- Designed and engineered an Assistive Eye Gaze Controller, achieving 95% real-time accuracy in translating eye movements into wheelchair commands using Blynk IoT API.
- Constructed a computer vision pipeline leveraging MediaPipe Face Mesh and Iris landmarks to track gaze position with sub-pixel precision, improving tracking accuracy by 30%.
- Implemented a control loop processing 30+ FPS real-time video frames to achieve command recognition latency under 2 seconds for mobility control.

Spectroplast: [GitHub](#)

Dec 2025 - Jan 2026

- Engineered a Physics-Informed Digital Twin to simulate Raman laser scattering, generating 1000+ synthetic hyperspectral cubes to data scarcity.
- Architected a U-Net 3D CNN to segment 10µm microplastics from 100-band spectral data, effectively filtering blood fluorescence.
- Deployed a Streamlit Clinical Dashboard integrating visual and chemical analysis to automate patient risk profiling and toxicity assessment.

ChromaShape: [GitHub](#)

Jun 2025 - Aug 2025

- Developed a web-based Personal Styling Assistant using Flask to provide data-driven clothing and style recommendations.
- Engineered a CV pipeline that automatically extracted 5+ unique features from user images, significantly enhancing personalization.
- Leveraged MediaPipe Pose Estimation and Feature Engineering to calculate the torso-to-leg ratio based on 10+ skeletal landmarks for comprehensive body proportion analysis.

AWARDS & ACHIEVEMENTS

- FAER Scholar Award Program 2025-26 – Shortlisted:** Shortlisted in the top 7% nationwide from 979 project proposals in a multidisciplinary engineering innovation challenge.
- Best First-Year Team – Codher'24 Hackathon:** Awarded Best First-Year Team by ACM-CEG, Anna University, solved real-time problem statements with a team of 3 and received a cash prize of Rs.1500.
- GeeksforGeeks:** Secured a score of 484 and achieved Institution Rank 12.

OPEN DATA CONTRIBUTIONS (Kaggle)

Kaggle Contributor – Datasets & Notebooks

• **Kaggle Profile:** kaggle.com/suryaprabha19

- Published structured datasets and end-to-end machine learning and data analytics notebooks covering EDA, feature engineering, model training, and evaluation.
- Awarded Kaggle Notebooks Bronze Medal for creating a high-quality ML pipeline with strong documentation and effective visualizations.